



Potential benefits of early-life supplementation of liquid feed with fennel (*Foeniculum vulgare*) seeds or oregano (*Origanum vulgare*) leaves on growth, health, and blood metabolites in Holstein dairy calves

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ABSTRACT

This study was performed to determine the early-life (first month of age) supplementation of liquid feed with fennel seed powder (FSP) or oregano leaf powder (OLP) on growth performance, health, and blood biochemical attributes in preweaning dairy calves. Holstein female calves ($n = 57$; 1 d of age; 34.1 ± 0.97 kg of BW; mean $\pm$ SE) were assigned randomly to receive liquid feed (colostrum and milk) with no added herbal plants (CON) or supplemented with FSP (3 g/d) or OLP (30 g/d) during the first month of age. The calves received pooled colostrum (4.5 kg/d on the first 2 d of life; total solids = $25.0\% \pm 1.24$; mean $\pm$ SD) and then pooled waste milk (6 kg/d from d 3 to 44, 5 kg/d from d 45 to 46, 4 kg/d from d 47 to 48, and 3 kg/d from d 49 to 50 of the trial; total solids = $12.54\% \pm 0.50$) to ensure they receive same mixed liquid feed daily. The calves had unlimited access to the starter feed and fresh water and remained in the study until weaning on d 51 of age. The average mean temperature-humidity index was 70.1 units (ranging between 61.9 to 78.2) during the experiment, indicating a borderline degree of environmental heat-load. The amount of starter feed offered and refused was recorded daily. The calves were weighed immediately after birth and every 10 d thereafter, before the morning feeding. Jugular blood samples were taken immediately before and 24 h after colostrum feeding, at first month of age, and at weaning to quantify serum concentrations of glucose, urea N, cholesterol, triglycerides, total proteins, albumin,

globulin, aspartate transferase, alanine transferase, total antioxidant status, and malondialdehyde. Health checks including rectal temperature, general appearance (on a 1–5 score system), fecal score (on a 1–5 score system), and bovine respiratory disease (BRD; scored using the University of Wisconsin Calf Health Chart) were performed daily, by a veterinarian who was unaware of the calf treatment allocations, for all calves over the study period. A repeated-measures ANOVA was used to compare growth performance and blood metabolites among treatment groups, and a logistic regression model using a binomial distribution (PROC GLIMMIX, SAS v. 9.4, SAS Institute Inc.) was used to assess the chance of elevated rectal temperature ($\geq 39.4^\circ\text{C}$), general appearance (≥ 2), diarrhea (≥ 3), and BRD. A Poisson regression model (PROC GENMOD) was also used to test group differences in the experience of days with elevated rectal temperature and general appearance, and frequency and duration of diarrhea or BRD. Total nutrient intake (DM, CP, and ME, but not ether extract) originating from milk and starter feed was greater in OLP- (but not FSP-) supplemented calves compared with CON group, being partially associated with difference in milk refusal. Calves receiving FSP and OLP had greater average daily gain, BW gain, skeletal gain (withers height or heart girth, respectively), and feed efficiency compared with CON animals with no difference between FSP and OLP. Rectal temperature was lower in FSP- (but not OLP-) supplemented calves compared with CON animals. The CON animals had a greater chance of experiencing higher rectal temperature ($\geq 39.4^\circ\text{C}$; odds ratio = 1.55 and confidence interval = 1.12–2.15 and odds ratio = 1.33 and confidence interval = 0.92–1.90, respectively, compared with FSP and OLP) and general appearance (≥ 2 ; odds ratio = 1.99 and confidence interval = 1.45–2.74 and odds ratio = 1.45 and confi-

Received January 2, 2022.

Accepted April 9, 2022.

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dence interval = 1.03–2.05), and diarrhea (odds ratio = 1.47 and confidence interval = 1.11–1.97 and odds ratio = 1.49 and confidence interval = 1.07–2.08) compared with those receiving FSP or OLP; with the chance of BRD being lower in FSP- (but not OLP-) supplemented versus CON animals (odds ratio = 1.59 and confidence interval = 1.13–2.23). As compared with OLP and CON groups, FSP treatment resulted in shorter days with elevated rectal temperature and general appearance. Supplementing FSP or OLP decreased the duration of diarrhea and BRD but not their frequency compared with CON. Duration of diarrhea was not different between FSP- or OLP-supplemented calves but calves supplemented with FSP had shorter days with BRD compared with OLP-supplemented calves. Of the blood constituents measured on d 30, concentration of aspartate transferase was higher in FSP- (but not CON) versus OLP-supplemented calves, indicating a transient liver tissue damage or dysfunction which was subsequently ameliorated. At weaning, blood concentration of triglycerides was higher in FSP and OLP groups compared with CON group. Supplementation of the liquid feed with FSP or OLP, especially FSP, had beneficial effects on calf growth performance and health. Further experiments are warranted for optimizing the dosage and duration of feeding FSP or OLP as feed additives for dairy calves.

Key words: heifer raising, herbal plant, summer, well-being

INTRODUCTION

Dairy calves are susceptible to digestive (56.0%) and respiratory (33.4%) disturbances during the milk feeding period that are costly to medicate and may also cause early mortality (32.0 and 14.1%, due to digestive and respiratory disturbances, out of all deaths; Urie et al., 2018). As reported by the USDA (2016), 33.8% of female dairy calves had experienced at least one morbidity event with 5% mortality rate during the milk feeding period. Therefore, morbidity and mortality rates during the milk feeding period exceed the Dairy Calf and Heifer Association (2010) targets of less than 25 and 5%, respectively. Accordingly, there is a need to substantially decrease the morbidity and mortality in dairy calves during the milk feeding period. As a preventive measure, conventional antibiotics are added to liquid or starter feeds, which may also lead to increased feed consumption and growth; however, the use of conventional antibiotics in animal-agriculture settings has been banned due to public health concerns and antibiotic resistance (Hajalizadeh et al., 2019). Therefore, there has been growing interest in the use of herbal plants such as fennel (*Foeniculum vulgare*; Saeedi et

al., 2017; Kargar et al., 2021; Nowroozinia et al., 2022) and oregano (*Origanum vulgare*; Selvi and Tapki, 2019; Tapki et al., 2020) as nutraceutical products in stimulating the growth and health of young animals.

The beneficial effects of these plants have been ascribed to their bioactive components with appetite stimulant properties mediated by essential oils, mainly *trans*-anethole in fennel (Saeedi et al., 2017; Hajalizadeh et al., 2019; Kargar et al., 2021) and carvacrol in oregano (Lei et al., 2018; Bosco Stivanin et al., 2019; Tapki et al., 2020), immune enhancing activity (Hashemipour et al., 2013; Badgujar et al., 2014), hepato-protective effects, hypoglycemic and hypolipidemic effects, and antioxidant and anti-inflammatory potential (due to compounds such as essential oils, flavonoids, phenolic acids, and others), antimicrobial properties (due to oleic acid, linoleic acid, 1,3-benzenediol, undecanal, 2,4-undecadienal, and 5-hydroxy-furanocoumarin in fennel and carvacrol in oregano; Badgujar et al., 2014; Hashemipour et al., 2013; Ri et al., 2017), and modulation of the digestive processes (Hashemipour et al., 2013; Saeedi et al., 2017; Wu et al., 2021).

Dose-dependent beneficial effects of dietary fennel seed powder (**FSP**) and oregano leaf powder (**OLP**) were reported in several species. These herbal plants have been used for treatment or prevention of the digestive and respiratory diseases (Badgujar et al., 2014; Alagawany et al., 2020; Nowroozinia et al., 2022), which are the most common diseases in preweaning calves (Urie et al., 2018). Supplementing the waste milk (during the first month of life) and starter feed (from first month of life until weaning) of Holstein dairy calves with FSP [3 g/d (but not 1.5 g/d) vs. control group (**CON**)] increased the starter feed intake, feed efficiency, weight gain, and skeletal growth (heart girth and hip width), and decreased the susceptibility to diarrhea and bovine respiratory disease (**BRD**), and reduced the number of days with elevated rectal temperature (Kargar et al., 2021; Nowroozinia et al., 2022). Supplementing the starter feed [0.4% (but not 0.8%) dietary DM vs. **CON**] of Holstein dairy calves (Saeedi et al., 2017) or fattening diet [1.5% (but not 0.75%) dietary DM vs. **CON**] of Kermani lambs (Hajalizadeh et al., 2019) with FSP increased the feed intake, feed efficiency, and growth rate (Saeedi et al., 2017; Hajalizadeh et al., 2019), reduced the weaning age (Saeedi et al., 2017), and had hepato-protective effects (Hajalizadeh et al., 2019) as evidenced by decreasing blood concentrations of aspartate transferase (**AST**) and alanine transferase (**ALT**), suggesting that FSP might protect liver damage or prevent hepatic dysfunction. Supplementing the liquid feed with oregano essential oil [at 100 mg/L (but not 150 mg/L) vs. **CON**] increased the overall calf BW gain (but not weaning weight) and feed efficiency, and

reduced the weaning age and fecal score (Tapki et al., 2020).

Supplementation of the goat diet with a blend of oregano essential oils and cobalt (Rum-A-Fresh, Ralco Inc.; 52 and 91 mg/d vs. CON) increased the weight gain and feed efficiency but decreased blood total protein and albumin concentrations without affecting blood concentrations of globulin, AST, and ALT (Lei et al., 2018). Supplementing a blend of oregano essential oils and cobalt (Rum-A-Fresh, Ralco Inc.; 4 and 7 g/d vs. CON) increased weight gain, feed efficiency, head and rump widths, and blood concentrations of glucose, total protein, albumin, and immunoglobulins (IgA, IgM, and IgG) but decreased concentrations of blood cholesterol, triglycerides, and globulin without affecting feed intake and nutrient digestibility except for CP digestibility that was higher in supplemented lambs (Wu et al., 2021). Oregano leaf powder also decreased feed intake in a dose-response manner in dairy cows supplemented at the levels higher than 1 g/d per kg of BW compared with CON but ECM and feed efficiency remained unchanged (Hristov et al., 2013). However, some types of herbal plants or formulated herbal essential oils or extracts as nutraceutical products at given dosages or routes and duration of administration have no beneficial effects on calf health and performance (Pempek et al., 2018; Selvi and Tapki, 2019; Jahani-Azizabadi et al., 2022).

Beneficial effects of medicinal plants or their components have also been reported in poultry, AL-Sagan et al. (2020) found that dietary supplementation of FSP [3.2% (but not 1.6%) vs. CON] improved the feed intake and weight gain in heat-stressed Ross-308 broiler chickens without affecting the feed efficiency. In addition, supplementing the broiler chicken diet 1:1 blend of pure thymol and carvacrol (at 60, 100, and 200 mg/kg; Next Enhance 150 containing 50% active components) or with 150 mg/kg OLP (rich in thymol and carvacrol) increased the feed intake and weight gain, enhanced antioxidant/digestive enzyme activities, reduced blood and meat malondialdehyde (MDA) concentrations, and increased the total antioxidant status (TAS); the immune responses were also enhanced as shown by increases in the hypersensitivity response, total and IgG antisheep red blood cell titers, as well as decreased heterophil to lymphocyte ratio compared with CON (Hashemipour et al., 2013; Ri et al., 2017).

To our knowledge, there is limited information on the effect of feeding fennel seeds or oregano leaves on growth and health attributes in preweaning dairy calves. Therefore, the goal of the present trial was to study the growth performance, health indices, and blood biochemical attributes in newborn Holstein dairy calves receiving FSP (3 g/d) or OLP (30 g/d) in their

liquid feed during the first month of age. We hypothesized that supplementing the liquid feed with FSP or OLP would increase the weight gain and improve the health indicators in calves.

MATERIALS AND METHODS

The trial was performed from July 23, 2021, to September 22, 2021, on a dairy farm (Dam Gostaran-e-Kavir Dairy Farm, Yazd, Iran) in compliance with the Iranian Council of Animal Care (1995) recommendations.

Climatic Conditions, Animals, and Management

Air temperature and relative humidity in the calf house were recorded several times daily (model HTC-1; Temperature Humidity Meter, Xuzhou Sanhe Automatic Control Equipment Co., Ltd.), and temperature-humidity index (THI) was computed according to NRC (2001). The average birth weight of calves used in the present study may imply that the dry cows were also environmentally heat-loaded, which can have carry-over effects on calf BW and performance (Karimi et al., 2015). Female calves (between 30 and 38 kg at birth), without signs of diseases, physical disabilities, failure of suckling, and fever, and with 24-h blood total protein level ≥ 5.5 g/dL, good general appearance and vigor (Murray-Kerr et al., 2018) were selected for the trial. Fifty-seven calves ($n = 19$ per treatment; first d of age) were allotted to treatments balanced for BW (34.1 ± 0.97 kg BW; mean $\pm$ SE), blood total protein (6.0 ± 0.31 g/dL), and dam parity (2.7 ± 0.20), and then housed to individual pens ($2.0 \times 1.1 \times 1.2$ m; length $\times$ width $\times$ height) during the 10 succeeding days (~ 6 calves daily; ~ 2 calves/treatment per day). The pens were naturally ventilated, with sand bedding that was refreshed every other day, and manure was removed daily to keep the pens visibly clean and dry.

Animals were fed 4.5 kg of colostrum (containing $25.0\% \pm 1.24$ DM, $17.08\% \pm 0.54$ CP, and $6.60\% \pm 0.46$ fat on as-fed basis) in a nipple bottle within the first hour of life (2.5 kg), and again 12 h after the first feeding (2.0 kg). On the second day of life, each calf was fed 4.5 kg/d colostrum in plastic buckets in 3 meals of equal volume (at 0400, 1200, and 2000 h). From the third day of life, the calves were individually fed with waste milk [nonsalable milk (temperature set at $39 \pm 1.0^\circ\text{C}$) containing low-quality colostrum (Brix value < 22), transition milk (the next 2 to 6 milkings after parturition), and drug residues; at 6 kg/d from d 3 to 44, 5 kg/d from d 45 to 46, 4 kg/d from d 47 to 48, and 3 kg/d from d 49 to 50 of the trial; containing $12.54\% \pm 0.50$ DM, $3.56\% \pm 0.12$ CP, and $3.62\% \pm 0.22$ crude

fat on as-fed basis] in plastic buckets in 3 meals of equal volume (at 0400, 1200, and 2000 h). To ensure consistent amount of liquid received by each calf, colostrum and waste milk were weighed at each feeding time. The farm-produced waste milk was transported daily to a refrigerated bulk tank equipped with a blender, and offered after blending; milk refusal was also recorded half an hour after each feeding time. The tank was daily flush-cleaned twice with tap water (at 20°C) for 5 min, and for the third time according to the following protocol: 5 min with tap water; 20 min with 75°C alkali solution containing 180 L of tap water and 2 kg of sodium hydroxide; 5 min with tap water; 20 min with 75°C acid solution containing 180 L of tap water and 3 L of sulfuric acid, and finally with tap water for 10 min.

FSP and OLP Dosage Selection and Treatments

Herbal plants likely affect the health and performance in a dose-response manner. The FSP dosage was chosen from our previous data on calves in which 3 g/d FSP significantly improved the growth performance and health-related variables (Kargar et al., 2021; Nowroozinia et al., 2022). Regarding the OLP dosage, we did not locate any research directly addressing the optimum OLP dosage for calves. However, there are reports on supplementation of OLP essential oils or extract without reporting the extraction efficiency to extrapolate the amount of OLP supplemented as essential oils/extract. In a study, Hristov et al. (2013) fed graded levels of OLP (0, 0.35, 0.70, and 1.05 g OLP/d per kg BW) in cows and found reduced feed intake when OLP dosage reached 1.05 without affecting the ECM and feed efficiency. Accordingly, we used a dosage between 0.70 and 1.05 g/d per kg of BW (means 0.88 g OLP/d per kg of calf BW, which corresponds to 30 g/d of OLP per calf, assuming the mean calf's BW of 34 kg). Therefore, the calves were assigned randomly to receive 3 g/d FSP or 30 g/d OLP (ground to pass through 1-mm screen) in liquid feed (noon feeding) during the first month of life, and weaned at d 51 of age. The CON did not receive any supplements. To ensure that each calf consumed the allocated amounts of herbal supplements, FSP or OLP were mixed with 250 mL of milk and fed in bottles. The remaining milk was fed according to the milk feeding program. Calves in CON group received their milk meal without supplemental herbs. The animals had also unlimited access to fresh and clean tap water and starter feed (mashed form; containing chopped alfalfa hay as sole source of forage at 7% of starter feed DM) offered in metal buckets (Table 1). The calf attendant was aware of the treatment allocation at the different stages of the trial (during the allocation, execution, and outcome as-

Table 1. Ingredients and chemical composition (% of DM unless otherwise noted) of the experimental diet

Composition	Total
Ingredient	
Alfalfa hay	7.0
Wheat bran	7.4
Corn grain, ground	37.0
Barley grain, ground	4.6
Soybean meal	34.0
Meat meal	2.7
Calcium carbonate	2.0
Sodium bicarbonate	1.5
Vitamin supplement ¹	1.5
Mineral supplement ²	1.5
Salt	0.5
Calcium phosphate (di-)	0.3
Chemical composition	
DM	89.9
CP	24.8
NFC ³	46.0
NDF	15.6
Ether extract	3.0
Ash	10.6
Ca	1.35
P	0.65
ME, ⁴ Mcal/kg of DM	2.97
Net energy for maintenance, ⁴ Mcal/kg of DM	2.22
Net energy for growth, ⁴ Mcal/kg of DM	1.69

¹Contained per kilogram of the supplement: 1,000,000 IU of vitamin A, 150,000 IU of vitamin D₃, 10,000 IU of vitamin E, 3.0 g of monensin, and 100 mg of biotin.

²Contained per kilogram of the supplement: 67.5 g of Mg, 21.5 g of Zn, 12.0 g of Mn, 5.4 g of Cu, 200 mg of Co, 100 mg of I, and 80 mg of Se.

³NFC = 100 - (CP + NDF + ether extract + ash) (NRC, 2001).

⁴Calculated from NRC (2001).

essment). The attendant only helped with the feeding, box cleaning, and other common husbandry duties but was not involved in any procedure such as information collection and interpretation that would affect the data reliability.

Sampling and Laboratory Analyses

The amounts of liquid and starter feeds (offered at 0800 h) and refusals were recorded daily for computation of the total feed intake. Starter feed (n = 6; pooled within trial period) and refusals (n = 5/calf; pooled by calf within treatment) were sampled immediately before the morning feeding every 10 d during the trial and stored at -20°C for chemical analysis. The DM content of samples was measured using a forced-air oven (100°C for 24 h; AOAC International, 2002; method 925.40). Samples, after thawing and mixing, were ground to pass a 1-mm screen (Wiley mill; Ogawa Seiki Co. Ltd.) and analyzed in triplicate for CP (Kjeltec 1030 Auto Analyzer, Tecator; AOAC International, 2002; method 955.04), ether extract (EE; AOAC International, 2002; method 920.39), crude ash (AOAC International, 2002;

Table 2. Chemical components based on the total essential oil (%) distilled from fennel (*Foeniculum vulgare*) seed or oregano (*Origanum vulgare*) leaf (n = 4 per herbal plant)

Fennel seed			Oregano leaf		
Compound	RI ¹	%	Compound	RI	%
(<i>E</i>)-Anethole (<i>trans</i> -anethole)	1,286	72.3	Carvacrol	1,298	26.5
Fenchone	1,088	9.9	<i>trans</i> -Sabinene hydrate	1,099	9.3
Limonene	1,030	6.1	Terpinen-4-ol	1,172	8.0
Estragole (methyl chavicol)	1,197	5.0	γ -Terpinene	1,058	6.7
α -Pinene	933	1.1	Terpinen-4-ol acetate	1,301	5.2
(<i>Z</i>)- β -Ocimene	1,035	0.9	(<i>E</i>)-Caryophyllene	1,419	4.6
Carvone	1,243	0.6	(<i>E</i>)-Anethole	1,286	4.0
Myrcene	990	0.6	α -Terpinene	1,017	3.4
<i>p</i> -Anisaldehyde, dimethyl acetal	1,254	0.5	<i>p</i> -Cymene	1,025	3.1
1,8-Cineole	1,031	0.4	<i>cis</i> -Sabinene hydrate	1,070	3.0
Sabinene	974	0.4	α -Terpineol	1,189	2.0
Camphor	1,144	0.3	β -Eudesmol	1,650	1.9
(<i>Z</i>)-Anethole (<i>cis</i> -anethole)	1,247	0.2	α -Eudesmol	1,654	1.7
α -Phellandrene	1,006	0.2	Terpinolene	1,089	1.6
Camphene	951	0.2	Carvacrol, methyl ether	1,245	1.4
γ -Terpinene	1,058	0.2	Borneol	1,169	1.3
<i>p</i> -Cymene	1,025	0.2	Caryophyllene oxide	1,582	1.2
Germacrene D	1,482	0.2	Limonene	1,030	1.2
exo-Fenchyl acetate	1,235	0.1	Sabinene	974	1.1
β -Pinene	978	0.1	Spathulenol	1,578	1.0
allo-Ocimene	1,127	0.1	Elemol	1,549	0.9
Carvacrol	1,298	0.1	Bicyclogermacrene	1,495	0.9
Terpinen-4-ol	1,172	0.1	<i>cis</i> - <i>p</i> -Menth-2-en-1-ol	1,124	0.9
Others	—	0.3	Others	—	8.6
Unknown	—	0.1	Unknown	—	0.3

¹Retention index relative to *n*-alkanes on column.

method 942.05), and NDF using a heat-stable α -amylase (100 μ L/0.5 g of sample) and sodium sulfite (Van Soest et al., 1991). The NFC component was computed as 100 – (CP + NDF + EE + Ash) (NRC, 2001). Three samples per day (one sample at each feeding time) were taken from pooled waste milk, preserved with potassium dichromate, stored at 4°C, and submitted to the Central Milk Testing Laboratory of the farm for determination of DM, fat, and protein concentrations using an infrared analyzer (MilkoScan 134 BN; Foss Electric).

Fennel seeds and oregano leaves were bought from a commercial store supplying traditional medicinal herbs (Yazd, Iran), and identified and authenticated by A. Hatami, a plant taxonomist at Fars Agricultural and Natural Resources Research and Education Center (Shiraz, Iran). The voucher specimen for fennel seeds (7714) and oregano leaves (23601) were deposited in the herbarium. Representative samples of shade dried fennel seeds and oregano leaves were ground to pass through 1-mm screen (Wiley mill; Ogawa Seiki Co. Ltd.). The ground samples (30 g; n = 4) were subjected to hydrodistillation in 1,000 mL of distilled water for 3 h using a Clevenger-type apparatus (HeatingMantle; Sci FineTech Co.). After dehydration over anhydrous sodium sulfate (CAS #7757–82–6), the percentage of oil was calculated. The prepared oil was stored in sealed vials in the dark at 4°C, and then subjected to GC

and GC-MS analyses for identification of constituents (Kargar et al., 2021; Table 2).

Jugular blood samples were taken immediately before and 24 h after colostrum feeding, at first month of age, and at weaning in Vacutainer tubes (BD Vacutainer) containing spray-coated silica, and immediately placed on ice. Blood samples were centrifuged at 3,000 $\times g$ for 20 min at 4°C, and 1.5 mL of serum was pipetted into 2-mL cryotubes and stored at –20°C until analysis. Serum glucose (mg/dL; Pars Azmoon Co.), BUN (mg/dL; Pars Azmoon Co.), cholesterol (mg/dL; Pars Azmoon Co.), triglycerides (mg/dL; Pars Azmoon Co.), total proteins (g/dL; Pars Azmoon Co.), albumin (g/dL; Pars Azmoon Co.), AST (U/L; aspartate transferase or serum glutamic oxaloacetic transaminase; Pars Azmoon Co.), ALT (U/L; alanine transferase or serum glutamic pyruvate transaminase; Pars Azmoon Co.), and TAS (mmol/L; Randox Laboratories Ltd.) were determined automatically by using standard test kits on an AL-CYON 300i automatic analyzer (Abbott Laboratories Ltd.). The analyzer was calibrated, and controls were assayed daily according to the manufacturer's instructions to ensure acceptable assay performance. Blood serum concentration of MDA (nmol/mL), a degradation product of lipid peroxidation, was quantified by thiobarbituric acid reacting substances method according to Kargar et al. (2015). Concentration of blood

globulin (g/dL) was calculated by subtracting albumin from total protein concentrations. The intra- and inter-assay coefficients of variations were 0.55 and 1.25% for glucose, 3.07 and 4.27% for BUN, 0.90 and 1.91% for cholesterol, 1.03 and 1.67% for triglycerides, 1.94 and 2.14% for total protein, 1.07 and 1.44% for albumin, 0.75 and 1.50% for AST, 2.25 and 2.85% for ALT, 0.45 and 0.93% for TAS, and 3.30 and 4.24% for MDA, respectively.

Growth

A calibrated electronic scale (model WLC; Etemad Co.) was used to measure BW immediately after birth and every 10 d thereafter before the morning feeding. The ADG (g of BW gain/d) was computed as the difference between BW recorded at 10-d intervals divided by 10. Body frame size [including the heart girth (chest circumference), withers height (distance from base of the front feet to the withers), body barrel (circumference of the paunch before feeding), hip height (distance from the base of the rear feet to hook bones), and hip width (distance between the points of hook bones)] was measured immediately after birth and at the end of the trial (Nowroozinia et al., 2022), and the skeletal gain (d 1 to 51) was calculated accordingly.

Health

Daily health checks were performed based on animal's appetite and the desire to consume liquid and starter feeds, and by monitoring the rectal temperature and general appearance (Heinrichs et al., 2003) by a veterinarian who was not aware of the calf treatment allocations. The general appearance was scored daily as 1 = normal and alert; 2 = ears drooped; 3 = head and ears drooped, dull eyes, slightly lethargic; 4 = head and ears drooped, dull eyes, lethargic; and 5 = severely lethargic (Heinrichs et al., 2003). To detect BRD, the animals were scored daily using the UW Calf Health Chart (McGuirk and Peek, 2014), which assigns and sums the scores for nasal discharge, eye discharge, ear tilt, cough, and temperature. Before morning feeding, fecal score was checked daily for physical shape and consistency (1 = normal; 2 = soft to loose; 3 = loose to watery; 4 = watery, mucous, slightly bloody; and 5 = watery, mucous, and bloody; Heinrichs et al., 2003) while the animals were in their individual pens. All calves were checked for rectal temperature daily between 1000 and 1200 h, using a digital thermometer (model MT-402; Digital Flexible Thermometer, Hangzhou Sejoy Electronics & Instruments Co.) that was inserted into the rectum for approximately 1 min. The rectal temperature was categorized as the number of days

with temperature reading $\geq 39.4^{\circ}\text{C}$ (as fever threshold), and fecal score was categorized as the number of days with a score ≥ 3 . These categories were designated as days with abnormal rectal temperature and fecal score, respectively. Calves with fever, abnormal fecal score or cough were examined by the veterinarian (unaware of the treatments) to confirm diarrhea or BRD.

Calves with diarrhea or BRD were medicated following the standard procedures at the Dam Gostarane-Kavir Animal Husbandry (Yazd, Iran). To treat diarrhea, animals received a water-based oral rehydration salt solution [Damy Balance; containing 650 mg dextrose, 145 mg sodium chloride, 40 mg potassium chloride, and 165 mg tri-sodium citrate per g; 4 L/d per calf in 2 meals of equal volume (at 0800 and 1600 h)] for 2 consecutive days; Damyaran Arak Veterinary Pharmaceutical Co.) or water soluble ScourSTOP powder [containing 970 mg of kaolin, 20 mg pectin, and 10 mg neomycin sulfate per g; 0.5 L/d per calf in 2 meals of equal volume (at 0800 and 1400 h) for 2 consecutive days; Tolide Darouhai Dami Iran Co.], and neomycin sulfate [four 500-mg bolus (2 bolus per feeding at 0500 and 2000 h) per day per calf for 2 consecutive days; Tolide Darouhai Dami Co.]. Nonresponding animals were treated for 3 more days with ceftriaxone [Afaxon; containing 1 g ceftriaxone as sodium per vial; 2 i.m. injections during the first day (at 0900 and 2100 h) and an additional single dose injection on d 2 and 3 (at 0900 h); AFA Chemistry Pharmaceutical Co.], gentamicin (Gentamicin 5%; containing 50 mg gentamicin/mL; 3 mL/d per calf for 3 consecutive days; Razak Laboratories Co., Karaj, Iran), and vitamin AD₃E (containing 50,000 IU vitamin A, 10,000 IU vitamin D₃, and 20 mg vitamin E per mL; 5 mL/d per calf for 3 consecutive days; Nasr Pharmaceutical Co., Torbat-e-Jam, Iran) injection. Animals with higher dehydration rate also received intravenous liquid therapy using ringer solution [containing 860 mg of sodium chloride, 33 mg of calcium chloride, and 30 mg of potassium chloride per 100 mL; 0.5 L/d per calf for 2 consecutive days; Shahid Ghazi Pharmaceutical Co.] plus sodium bicarbonate (BicarboJect; containing 84 mg of sodium bicarbonate/mL; 100 mL/d per calf for 2 succeeding days; Nasr Pharmaceutical Co.). Animals with bloody diarrhea received enrofloxacin (Enrocin 5%; containing 50 mg enrofloxacin/mL; 5 mL/d per calf for 3 consecutive days; Razak Laboratories Co.), Pantrisol (containing 40 mg trimethoprim and 200 mg sulfamethoxazole/mL; 5 mL/d per calf for 3 consecutive days; Pantex Holland B.V.), and vitamin K (MenadioVit; containing 100 mg of menadione sodium bisulfite/mL; 5 mL/d per calf for 1 to 3 consecutive days depending on blood disappearance in feces; Nasr Pharmaceutical Co.) injection.

Animals were treated for BRD with bromhexine (BroncoJect; containing 3 mg bromhexine/mL; 5 mL/d per calf for 5 consecutive days; Rooyan Darou Pharmaceutical Co.), florfenicol (F-Nex 300; containing 300 mg florfenicol/mL; 5 mL/d per calf on the first 3 consecutive days; Razak Laboratories Co.), a single dose of dexamethasone (Dexalax 0.2%; containing 2.63 mg dexamethasone as sodium phosphate/mL; 2 mL per calf on the first day of treatment protocol; Bayer Aflak Pharma Co.), meloxicam (MeloxiVet 2%; containing 20 mg of meloxicam/mL; 5 mL/d per calf on the second and third day of treatment protocol; Razak Laboratories Co.), and vitamin AD₃E (5 mL/d per calf during the first 3 d; Nasr Pharmaceutical Co.) injection. Non-responding animals received aminophylline [Teolin; containing 25 mg aminophylline (theophylline anhydrous)/mL; 10 mL/d per calf (5 mL/injection in 12-h intervals) for 5 consecutive days; Teolin; Sina Darou], amikacin [Amikacin 250-Exir; containing 250 mg amikacin/mL; 4 mL/d per calf (2 mL/injection in 12-h intervals) for 3 consecutive days; Exir Pharmaceutical Co.), a single dose of dexamethasone (2 mL per calf on the first day of treatment protocol; Bayer Aflak Pharma Co.), meloxicam (5 mL/d per calf on the second and third day of treatment protocol; Razak Laboratories Co.), and vitamin AD₃E (5 mL/d per calf for 3 consecutive days; Nasr Pharmaceutical Co.) injection.

Statistical Analysis

A classical power analysis (POWER PROC; SAS, version 9.4; SAS Institute Inc.) was performed to determine the required number of experimental units. The power ($1 - \beta$) was chosen to be equal to 85%, and the α -level was 0.05. The power analysis was performed for ADG as one of the most reliable parameters to consider in calf performance. According to the outcome of a previous experiment performed by Nowroozinia et al. (2022) with 36 calves assigned into 3 treatment groups, a standard deviation of 95 g/d was assumed for ADG. The minimal meaningful difference in ADG was considered to be 105 g/d. The minimal sample size to detect differences at this assumption was computed to be 12 calves per treatment group. Accordingly, 57 calves were considered sufficient to get a significant result with adequate power in performance among the treatment groups. To the best of our knowledge, no study has addressed the potential of FSP and OLP in preventing diarrhea or BRD in dairy calves except Nowroozinia et al. (2022) who studied the effect of FSP on health events using 16 calves per treatment. Furthermore, because there were no data on true herd diarrhea or BRD, we did not carry out power analyses for the health-related variables. However, maximum number of calves

were used based on the potential available pens on the farm; this was a convenience sample.

Data were assessed for normality using the PROC UNIVARIATE in the SAS software (SAS Institute Inc.). The data that were not normally distributed were transformed logarithmically. Data on performance, rectal temperature, and blood biochemical parameters were subjected to ANOVA using the PROC MIXED with times (1- or 10-d period) as the repeated measures. Calf was considered as the random effect, and treatment (**T**; CON, FSP, and OLP), period (**P**; 1- or 10-d period) and $T \times P$ interaction as the fixed effects. Animals with diarrhea or BRD were considered sick; this dichotomous variable (sick or not sick during the trial) was also included in the model but was not significant and excluded from the final analysis. Data on initial BW and initial skeletal measurements and gains were analyzed using the same model without the period effect. Initial blood variables, initial BW, and initial body frame size were used as the covariates for blood biochemical parameters, BW, and skeletal size analysis, respectively. The ante-dependence covariance structure (type 1) was the best fit for these data as determined by the lowest value for the Bayesian information criterion. Data were reported as the least squares means, and considered significant if $P \leq 0.05$ and tendency if $0.05 < P \leq 0.10$. When the interaction between T and P was significant, the SLICE option in the LSMEANS statement was used to determine the differences among the factors.

Models for the occurrence of elevated rectal temperature ($\geq 39.4^\circ\text{C}$), general appearance (≥ 2), diarrhea (≥ 3), and BRD were assessed during the trial (d 1 to 51) by logistic regression using a binomial distribution in the PROC GLIMMIX. The odds ratio was used to compare the likelihood for calves on each treatment group to experience any event. Days with elevated rectal temperature and general appearance, and frequency and duration of diarrhea or BRD were assessed with a Poisson distribution using the PROC GENMOD command. No data were deposited in an official repository.

RESULTS

Climatic Conditions and Essential Oil Composition

The average means and ranges for THI, relative humidity, and temperature were 70.1 units (61.9–78.2 units), 25.7% (21.1–30.4%), and 26.4°C (18.3–34.5 $^\circ\text{C}$), respectively, during the experiment. The mean essential oil yield was 1.56 and 0.95% for fennel seed and oregano leaf, respectively. The fennel and oregano essential oils contained 39 and 55 compounds accounting for 99.9 and 99.7% of their total essential oils, respectively

Table 3. Nutrient intake (d 1–51) as influenced by early-life (first 30-d) supplementing liquid feed without (CON) or with fennel seed powder (FSP; 3 g/d) or oregano leaf powder (OLP; 30 g/d) to neonatal Holstein calves (n = 19 per treatment)

Item	Treatment (T) LSM			SEM	P-value		
	CON	FSP	OLP		T	Period (P)	T × P
Starter feed intake, g/d							
DM	118 ^b	129 ^{ab}	169 ^a	20.04	0.02	0.001	0.19
CP	29 ^b	32 ^{ab}	42 ^a	4.97	0.05	0.001	0.19
EE	4	4	5	0.60	0.22	0.001	0.20
ME, Mcal/d	0.35 ^b	0.38 ^{ab}	0.50 ^a	0.06	0.02	0.001	0.19
Total feed intake, ¹ g/d							
DM	846 ^b	856 ^{ab}	896 ^a	20.04	0.03	0.001	0.19
CP	215 ^b	218 ^{ab}	228 ^a	4.97	0.02	0.001	0.19
EE	186	186	187	0.60	0.22	0.001	0.20
ME, Mcal/d	4.06 ^b	4.09 ^{ab}	4.21 ^a	0.06	0.02	0.001	0.19

^{a,b}Means within a row for each effect with different superscripts are different ($P \leq 0.05$).

¹Total feed intake = milk intake + starter feed intake.

(Table 2). The fennel essential oil largely consisted of monoterpenes (99.6%; such as *cis*- and *trans*-anethole, fenchone, limonene, estragole, and α -pinene), followed by sesquiterpenes (0.31%; such as germacrene D and α -copaene). The oregano essential oil largely consisted of monoterpenes (79.7%; such as carvacrol, *trans*-sabinene hydrate, terpinen-4-ol, γ -terpinene, terpinen-4-ol acetate, *trans*-anethole, α -terpinene, *p*-cymene, *cis*-sabinene hydrate, α -terpineol, terpinolene, carvacrol (methyl ether), borneol, limonene, and sabinene), followed by sesquiterpenes (13.80%; such as *trans*-caryophyllene, β -eudesmol, α -eudesmol, caryophyllene oxide, and spathulenol). *Trans*-anethole (72.3%) and carvacrol (26.5%) were the main monoterpene and germacrene D (0.2%) and *trans*-caryophyllene (4.6%) as the main sesquiterpene compounds in fennel and oregano essential oils, respectively. The fennel and oregano essential oils also contained considerable amounts of various minor compounds with proportional contribution under 0.3 and 8.6%, respectively.

Growth Attributes

There was no interaction between treatment and period on the nutrient intake ($P \geq 0.19$), originating from starter feed intake or starter feed intake plus milk intake, which increased ($P = 0.001$) with advancing calf age (Table 3). Supplementing OLP, but not FSP, increased the intake of DM (LSM = 169 vs. 118 g/d; SEM = 20.04; $P = 0.02$), CP (LSM = 42 vs. 29 g/d; SEM = 4.97; $P = 0.05$), and ME (LSM = 0.50 vs. 0.35 Mcal/d; SEM = 0.06; $P = 0.02$) originating from the starter feed and starter plus liquid feeds compared with CON. The calves supplemented with FSP or OLP recorded increased BW on d 40 (LSM = 56.9 and 55.7 vs. 52.6 kg for FSP and OLP vs. CON; SEM = 1.30;

$P = 0.06$) and 50 (LSM = 63.7 and 63.9 vs. 59.4 kg for FSP and OLP vs. CON; SEM = 1.67; $P = 0.04$). Average daily gain (LSM = 602 and 600 vs. 492 g/d for FSP and OLP vs. CON, respectively; SEM = 30.30; $P = 0.03$), final BW (LSM = 63.7 and 63.9 vs. 59.4 kg; SEM = 1.67; $P = 0.01$), BW gain (d 1–51; LSM = 30.1 and 30.0 vs. 24.6 kg; SEM = 1.67; $P = 0.01$), and feed efficiency (LSM = 0.70 and 0.67 vs. 0.58; SEM = 0.03; $P = 0.04$) were greater in FSP- and OLP-supplemented calves than in CON animals (Table 4). The barrel size, hip height, and hip width were not different between treatment groups; however, withers height (LSM = 90.2 vs. 88.2 cm; SEM = 0.90; $P = 0.04$) and final heart girth (LSM = 94.1 vs. 92.0 cm; SEM = 0.73; $P = 0.02$) as well as their gains were higher in FSP- and OLP-supplemented calves, respectively, than in CON animals. Skeletal measurements were not different between FSP- or OLP-supplemented calves.

Health Indices

Supplementing FSP, but not OLP, decreased the rectal temperature compared with CON animals (LSM = 38.7 vs. 38.9°C in FSP vs. CON; SEM = 0.02; $P = 0.05$) (Table 4). Table 5 shows the logistic models for the occurrence of elevated rectal temperature ($\geq 39.4^\circ\text{C}$), general appearance score (≥ 2), diarrhea (≥ 3), and BRD. The CON calves had and tended to have a greater chance of having elevated rectal temperature than those supplemented with FSP (odds ratio = 1.55; confidence interval = 1.12–2.15; SEM = 0.16; $P = 0.007$) or OLP (odds ratio = 1.33; confidence interval = 0.92–1.90; SEM = 0.18; $P = 0.10$), respectively, with no difference between FSP- and OLP-supplemented calves. The CON calves had a greater chance of having a general appearance score

Table 4. Growth performance (d 1–51), respiration rate, and rectal temperature as influenced by early-life (first 30 d) supplementing liquid feed without (CON) or with fennel seed powder (FSP; 3 g/d) or oregano leaf powder (OLP; 30 g/d) to neonatal Holstein calves (n = 19 per treatment)

Item	Treatment (T) LSM			SEM	P-value		
	CON	FSP	OLP		T	Period (P)	T × P
ADG (d 1–51), g/d	492 ^b	602 ^a	600 ^a	30.30	0.03	0.001	0.31
BW, kg							
Initial (d 1)	34.8	33.6	33.9	0.97	0.63	—	—
Final (d 51)	59.4 ^b	63.7 ^a	63.9 ^a	1.67	0.01	—	—
Gain (d 1–51)	24.6 ^b	30.1 ^a	30.0 ^a	1.67	0.01	—	—
Overall (d 1–51)	47.7	50.2	49.4	1.17	0.28	0.001	0.04
Feed efficiency (d 1–51)							
ADG/total DM intake	0.58 ^b	0.70 ^a	0.67 ^a	0.03	0.04	0.001	0.24
ADG/total ME intake	0.12 ^b	0.15 ^a	0.14 ^a	0.007	0.03	0.001	0.30
Heart girth, cm							
Initial (d 1)	77.1	77.2	77.5	0.88	0.95	—	—
Final (d 51)	92.0 ^b	93.2 ^{ab}	94.1 ^a	0.73	0.02	—	—
Gain (d 1–51)	14.9 ^b	16.0 ^{ab}	16.6 ^a	0.73	0.04	—	—
Overall (d 1–51)	85.6 ^b	86.9 ^{ab}	87.8 ^a	0.62	0.03	0.001	0.84
Withers height, cm							
Initial (d 1)	76.8	76.4	77.0	1.12	0.92	—	—
Final (d 51)	88.2 ^b	90.2 ^a	88.7 ^{ab}	0.90	0.04	—	—
Gain (d 1–51)	11.4 ^b	13.8 ^a	11.7 ^{ab}	0.90	0.01	—	—
Overall (d 1–51)	83.0 ^b	84.5 ^a	83.9 ^{ab}	0.57	0.04	0.001	0.53
Body barrel, cm							
Initial (d 1)	75.0	74.1	74.9	0.98	0.76	—	—
Final (d 51)	89.4	90.6	90.9	1.02	0.23	—	—
Gain (d 1–51)	14.4	16.5	16.0	1.02	0.12	—	—
Overall (d 1–51)	83.8	84.2	84.5	0.70	0.83	0.001	0.72
Hip height, cm							
Initial (d 1)	78.2	78.4	77.4	1.01	0.80	—	—
Final (d 51)	90.4	91.0	91.0	0.89	0.28	—	—
Gain (d 1–51)	12.2	12.6	13.6	0.89	0.22	—	—
Overall (d 1–51)	85.1	85.6	85.8	0.67	0.70	0.001	0.70
Hip width, cm							
Initial (d 1)	16.9	17.0	16.1	0.37	0.16	—	—
Final (d 51)	23.4	23.6	22.9	0.29	0.26	—	—
Gain (d 1–51)	6.5	6.6	6.8	0.29	0.26	—	—
Overall (d 1–51)	20.6	21.0	20.9	0.22	0.33	0.001	0.18
Rectal temperature, °C	38.9 ^a	38.7 ^b	38.8 ^{ab}	0.02	0.05	0.001	0.28

^{a,b}Means within a row for each effect with different superscripts are different ($P \leq 0.05$).

(≥ 2) for the duration of the study than FSP (odds ratio = 1.99; confidence interval = 1.45–2.74; SEM = 0.16; $P = 0.001$) or OLP (odds ratio = 1.45; confidence interval = 1.03–2.05; SEM = 0.17; $P = 0.03$) calves with no difference between FSP- and OLP-supplemented calves. The CON calves had a greater chance of having diarrhea (odds ratio = 1.47; confidence interval = 1.11–1.97; SEM = 0.14; $P = 0.007$) and BRD (odds ratio = 1.59; confidence interval = 1.13–2.23; SEM = 0.17; $P = 0.007$) compared with FSP-supplemented calves. The occurrence of diarrhea (odds ratio = 1.49; confidence interval = 1.07–2.08; SEM = 0.17; $P = 0.01$), but not BRD (odds ratio = 1.26; confidence interval = 0.87–1.83; SEM = 0.18; $P = 0.20$), was greater in CON calves than those supplemented with OLP. The chance of diarrhea (odds ratio = 1.01; confidence interval = 0.71–1.43; SEM = 0.17; $P = 0.95$) or BRD (odds ratio = 0.79; confidence in-

terval = 0.53–1.18; SEM = 0.20; $P = 0.26$) was not different between FSP- or OLP-supplemented calves.

Table 6 shows the Poisson regression for days (modeled means as LSM) with elevated rectal temperature ($\geq 39.4^\circ\text{C}$), general appearance score (≥ 2), and frequency and duration of diarrhea (≥ 3) or BRD during the study period. The CON calves recorded more days with elevated rectal temperature (SEM = 0.12; $P = 0.03$) and general appearance score (≥ 2 ; SEM = 0.11; $P = 0.001$) compared with the FSP- (2.0 and 3.4 d) and OLP- (1.4 and 2.1 d) supplemented calves. Also, OLP-supplemented calves experienced more days with elevated rectal temperature (0.6 d; SEM = 0.12; $P = 0.03$) and general appearance score (1.3 d; SEM = 0.11; $P = 0.001$) compared with FSP-supplemented animals. The frequency of diarrhea or BRD was not affected by treatment; however, diarrhea lasted more days in CON calves than those supplemented with FSP (2.3 d; SEM

Table 5. Logistic model for the occurrence of elevated rectal temperature ($\geq 39.4^\circ\text{C}$), general appearance (≥ 2)¹, diarrhea (≥ 3)², and bovine respiratory disease (BRD) occurrence during the experiment (d 1–51) as influenced by early-life (first 30 d) supplementing liquid feed without (CON) or with fennel seed powder (FSP; 3 g/d) or oregano leaf powder (OLP; 30 g/d) to neonatal Holstein calves (n = 19 per treatment)

Variable and comparison	Coefficient	SEM	OR ³	95% CI	P-value
Rectal temperature					
CON vs. FSP	0.4417	0.16	1.55	1.12, 2.15	0.007
CON vs. OLP	0.2851	0.18	1.33	0.92, 1.90	0.10
FSP vs. OLP	-0.1566	0.19	0.85	0.58, 1.25	0.42
General appearance					
CON vs. FSP	0.6909	0.16	1.99	1.45, 2.74	0.001
CON vs. OLP	0.3767	0.17	1.45	1.03, 2.05	0.03
FSP vs. OLP	-0.3141	0.19	0.73	0.50, 1.06	0.11
Diarrhea occurrence					
CON vs. FSP	0.3917	0.14	1.47	1.11, 1.97	0.007
CON vs. OLP	0.4016	0.17	1.49	1.07, 2.08	0.01
FSP vs. OLP	0.0098	0.17	1.01	0.71, 1.43	0.95
BRD occurrence					
CON vs. FSP	0.4660	0.17	1.59	1.13, 2.23	0.007
CON vs. OLP	0.2373	0.18	1.26	0.87, 1.83	0.20
FSP vs. OLP	-0.2287	0.20	0.79	0.53, 1.18	0.26

¹Where 1 = normal and alert; 2 = ears drooped; 3 = head and ears drooped, dull eyes, slightly lethargic; 4 = head and ears drooped, dull eyes, lethargic; and 5 = severely lethargic (Heinrichs et al., 2003).

²Where 1 = normal; 2 = soft to loose; 3 = loose to watery; 4 = watery, mucous, and slightly bloody; and 5 = watery, mucous, and bloody (Heinrichs et al., 2003).

³The odds ratio (OR) indicates the probability of either having elevated rectal temperature ($\geq 39.4^\circ\text{C}$), general appearance (≥ 2), diarrhea (≥ 3), or BRD for the experimental diets (CON vs. FSP; CON vs. OLP; and FSP vs. OLP). If the OR is > 1 , a given diet in the comparison is more likely to have elevated rectal temperature ($\geq 39.4^\circ\text{C}$), general appearance (≥ 2), diarrhea (≥ 3), or BRD than the other diet by a factor of the difference above 1. If the OR is < 1 , a given diet has a lower probability of occurrence than the other diet.

= 0.10; $P = 0.01$) or OLP (2.4 d; SEM = 0.10; $P = 0.01$). Duration of diarrhea was not different between FSP- or OLP-supplemented calves. The CON calves experienced more days with BRD compared with FSP- (2.0 d; SEM = 0.12; $P = 0.03$) and OLP- (1.1 d; SEM = 0.12; $P = 0.03$) supplemented animals. Furthermore, OLP-supplemented calves experienced longer days with

BRD compared with FSP-supplemented animals (1.1 d; SEM = 0.12; $P = 0.03$).

Blood Biochemical Measurements

The CON calves had lower serum concentration of triglycerides at weaning compared with those supple-

Table 6. Poisson regression for days with elevated rectal temperature ($\geq 39.4^\circ\text{C}$) and general appearance (≥ 2)¹, and frequency and duration of diarrhea (≥ 3)² and bovine respiratory disease (BRD) during the experiment (d 1–51) as influenced by early-life (first 30 d) supplementing liquid feed without (CON) or with fennel seed powder (FSP; 3 g/d) or oregano leaf powder (OLP; 30 g/d) to neonatal Holstein calves (n = 19 per treatment)

Item	Treatment (T) LSM				P-value
	CON	FSP	OLP	SEM	T
Days with elevated rectal temperature ($\geq 39.4^\circ\text{C}$)	6.3 ^c	4.3 ^a	4.9 ^b	0.12	0.03
Days with elevated general appearance (≥ 2)	7.5 ^c	4.1 ^a	5.4 ^b	0.11	0.001
Diarrhea					
Frequency, no. of times diagnosed	2.2	2.3	2.0	0.18	0.87
Duration, d	8.1 ^b	5.8 ^a	5.7 ^a	0.10	0.01
BRD					
Frequency, no. of times diagnosed	1.0	0.6	0.9	0.30	0.37
Duration, d	5.8 ^c	3.8 ^a	4.7 ^b	0.12	0.03

^{a-c}Means within a row for each effect with different superscripts are different ($P \leq 0.05$).

¹Where 1 = normal and alert; 2 = ears drooped; 3 = head and ears drooped, dull eyes, slightly lethargic; 4 = head and ears drooped, dull eyes, lethargic; and 5 = severely lethargic (Heinrichs et al., 2003).

²Where 1 = normal; 2 = soft to loose; 3 = loose to watery; 4 = watery, mucous, and slightly bloody; and 5 = watery, mucous, and bloody (Heinrichs et al., 2003).

Table 7. Blood serum metabolic, enzymatic, and antioxidant parameters as influenced by early-life (first 30-d) supplementing liquid feed without (CON) or with fennel seed powder (FSP; 3 g/d) or oregano leaf powder (OLP; 30 g/d) to neonatal Holstein calves (n = 19 per treatment)

Item	Treatment (T) LSM				P-value		
	CON	FSP	OLP	SEM	T	Period (P)	T × P
Metabolic parameter							
Glucose, mg/dL	78.70	78.28	82.51	6.31	0.87	0.07	0.25
BUN, mg/dL	21.41	22.33	20.97	1.27	0.74	0.65	0.85
Cholesterol, mg/dL	73.04	78.79	78.25	3.30	0.41	0.001	0.39
Triglyceride, mg/dL	30.20 ^b	39.72 ^a	37.00 ^a	2.72	0.05	0.02	0.02
Total protein, g/dL	5.39	5.67	5.56	0.21	0.67	0.007	0.74
Albumin, g/dL	3.26	3.43	3.38	0.06	0.22	0.001	0.61
Globulin, g/dL	2.17	2.21	2.18	0.19	0.98	0.001	0.57
Albumin:globulin	1.97	1.97	1.92	0.21	0.97	0.001	0.36
Enzymatic parameter							
Aspartate transferase, U/L	72.02	73.36	74.99	3.99	0.86	0.001	0.01
Alanine transferase, U/L	13.87	14.35	16.50	1.44	0.39	0.001	0.78
Antioxidant parameters							
Total antioxidant status, mmol/L	0.54	0.58	0.60	0.03	0.42	0.001	0.60
Malondialdehyde, nmol/mL	2.66	2.62	2.71	0.17	0.93	0.001	0.65

^{a,b}Means within a row for each effect with different superscripts are different ($P \leq 0.05$).

mented with FSP or OLP (LSM = 21.46 vs. 43.52 or 38.02 mg/dL for CON vs. FSP or OLP, respectively; SEM = 3.63; $T \times P = 0.02$) (Table 7) with no difference between FSP- or OLP-supplemented calves. A significant treatment by period interaction showed the FSP-supplemented calves had greater concentration of AST on d 30 compared with those OLP-supplemented calves (LSM = 63.77 vs. 43.51 U/L for FSP vs. OLP; SEM = 4.41; $T \times P = 0.01$); whereas there was no difference between the CON (LSM = 56.78 U/L) and other groups. Other blood constituents were not affected by treatment (Table 7).

DISCUSSION

The yield of hydrodistilled fennel seed or oregano leaf essential oils in the present study was in line with the values reported in the literature. *Trans*-anethole and carvacrol are the major components of essential oils from fennel seeds or oregano leaves native to Iran but the relative value of chemical compounds varies considerably depending on the origin, phonological state, agro-climatic conditions, and plant adaptive metabolism (Morshedloo et al., 2018). *Trans*-anethole and carvacrol play a key role in the overall aroma of fennel or oregano, respectively, and therefore, affect the mechanisms of action of the essential oil compounds (Morshedloo et al., 2018; Tapki et al., 2020; Kargar et al., 2021).

The average maximum THI was 78.2 during the experiment, indicating a high degree of heat-load. In central Iran, THI is often above 78 from May through September (Kargar et al., 2018a,b). The THI of 65 to 69 is considered the threshold cut-point for dairy calves

(Dado-Senn et al., 2020b) above which, hyperthermia negatively affects the calf performance (Kargar et al., 2018a,b; Dado-Senn et al., 2020a) unless heat stress abatement protocol as well as nutritional and managerial interventions are implemented. However, the overall average THI was 70.1, which is on the borderline of heat stress for calves, whereas the minimum THI (61.9) was much below this threshold level indicating that calves had not been chronically experienced heat stress, and may have had sufficient time to recover their feed intake, thus limiting the effect of supplements to stimulate their appetite. Therefore, possibly the lack of effect on stimulating appetite which was not observed in these calves is because that the mean THI in the calf barn was borderline/not environmental conditions of heat stress and there were plenty of times every day that the ambient environment was well below the THI threshold for heat stress in calves. Furthermore, calves born to environmentally heat-loaded dams, as was the case in the present study, usually have lower birth weight (Karimi et al., 2015), and thereby experience a moderate level of weight gain.

It was reported earlier that FSP had the potential to increase feed intake through decreasing the time to initiate the first meal after starter feed provision by increasing the oro-sensorial preference, reducing the non-nutritive oral behaviors as well as increasing the eating rate and meal size in milk-fed calves (Saeedi et al., 2017; Kargar et al., 2021) and fattening lambs (Hajalizadeh et al., 2019). The nonsignificant effects of FSP on feed intake in the present study might be attributed to the method of FSP application [only in liquid feed in the present study compared with in liquid plus starter feeds (Kargar et al., 2021) or in starter

feed only (Saeedi et al., 2017; Hajalizadeh et al., 2019)], supplementation duration [first month of life in the present study vs. whole preweaning or fattening periods (Saeedi et al., 2017; Hajalizadeh et al., 2019; Kargar et al., 2021)], ambient temperature [hyperthermia in the present study vs. thermoneutral conditions in previous studies (Saeedi et al., 2017; Hajalizadeh et al., 2019; Kargar et al., 2021)], and supplementation level. The present supplementation level was chosen based on the data from the experiments performed in thermoneutral conditions (Kargar et al., 2021; Nowroozinia et al., 2022) and needs to be re-evaluated for experimentation under heat-loaded conditions. Alternatively, we did not offer enough milk to the calves to see this “stimulatory effect” because calves would have consumed more milk than was offered per day in the present study.

Essential oils originating from plants may affect the neuronal activity via modulation of neurotransmitters (such as serotonin and dopamine), which potentially modify the animal’s response to the environment (Trabace et al., 2011; Zotti et al., 2013). In parallel research in rats, it was shown that carvacrol crossed the blood–brain barrier and acted on the feeding center in the central nervous system by decreasing the 5-HT receptors (Trabace et al., 2011; Zotti et al., 2013), serotonin and dopamine (Trabace et al., 2011; Zotti et al., 2013), and stimulated the feeding behavior (Simansky, 1995; Lucki, 1998). This property of carvacrol may explain the increase in starter and total nutrient intakes before weaning, although neither the concentration of 5-HT receptors nor serotonin and dopamine were measured in the present trial. The literature on the effects of the OLP or oregano essential oil/extract on feed intake is contradictory. Consistent with the results of the present study, supplementing the diet with OLP or oregano essential oil/extract increased the feed intake in milk-fed calves (Tapki et al., 2020), broiler chickens (Ri et al., 2017), goats (Lei et al., 2018), and dairy cows (Bosco Stivanin et al., 2019). However, decreased feed intake or nonsignificant effects on intake were also reported (Bampidis et al., 2005; Chaves et al., 2008; Hristov et al., 2013; Kolling et al., 2016, 2018; Wu et al., 2021), indicating the need for further clarification of the conditions for the use of FSP or OLP in various animal species. Overall, the observed intake of starter feed was substantially lower than when liquid feed intake is restricted, and current values are well in line with other studies (e.g., Berends et al., 2020).

The beneficial effects of FSP on the weight gain and body or frame size, also reported earlier in dairy calves (Saeedi et al., 2017; Kargar et al., 2021; Nowroozinia et al., 2022) and fattening lambs (Hajalizadeh et al., 2019), may, in part, be related to higher feed efficiency due to estrogenic compounds (such as *trans*-anethole) in fennel

seeds. Estrogenic compounds also play important roles in skeletal muscle and bone growth (Nowroozinia et al., 2022), increase glucose uptake into muscle cells and provide long-lasting anabolic conditions (Saeedi et al., 2017; Hajalizadeh et al., 2019), modulate the immune responses, and are anti-inflammatory (Badgujar et al., 2014; Nowroozinia et al., 2022). The stimulating effect of OLP on ADG and frame size may, in part, be attributed to the increased feed intake stimulated by the carvacrol, the main component of oregano essential oils (Ri et al., 2017; Lei et al., 2018; Tapki et al., 2020), as well as the higher feed efficiency (Lei et al., 2018; Wu et al., 2021). Dietary supplementation of carvacrol (at 0.5 and 2 g/kg DM) improved gut health and absorption efficiency in postweaning piglets through decreasing the number of intra-epithelial lymphocytes and increasing the ratio of villus height to crypt depth in the small intestine (Michiels et al., 2010).

Rectal temperature was in the physiological range (38.1–39.2°C; Kargar et al., 2018a,b; Heisler et al., 2020), indicating a reasonable calf ability to cope with the potential thermal stress in the present study, especially in FSP-supplemented calves that recorded lower rectal temperature. It has been shown that feeding oregano extract to Holstein heifers (at 2.5, 5.0, and 7.5 g/d; Kolling et al., 2016) or preweaning Jersey calves (at 60 mg/kg of BW per day; Heisler et al., 2020) did not affect rectal temperature under thermoneutral conditions; however, to the best of our knowledge, similar data are not available for comparison with our findings. Nowroozinia et al. (2022) reported lower susceptibility to having elevated rectal temperature and fewer days with elevated rectal temperature when milk-fed calves received FSP at 1.5 or 3 g/d.

The beneficial effects of FSP and OLP on diarrhea and BRD can be attributed to their well-established medicinal properties in humans, which have not all been investigated fully in calves. Therefore, interpretation to calves must be done with caution. Fennel has been shown to be effective in treatment of gastrointestinal (including diarrhea) and respiratory disturbances (Badgujar et al., 2014; Nowroozinia et al., 2022). Also, oregano has the potential to reduce the digestive and respiratory disturbances in some species (Games et al., 2016; Gilling et al., 2019; Alagawany et al., 2020). Therefore, improvement in the gut health may decrease diarrhea and related health events in calves (Tapki et al., 2020). It has been shown that when calves suffering from diarrhea due to *Escherichia coli* were treated with OLP (providing 10 mg oregano essential oil/kg of BW per calf per day), their fecal score, severity of diarrhea, and mortality rate decreased (Bampidis et al., 2006). Also, the calves supplemented with oregano essential oil (at 12.5 mg/kg of BW per day) for the first 10 d

of life experienced a decreased incidence of diarrhea as well as duration and severity of the disease compared with unsupplemented calves (Katsoulos et al., 2017). For comparative purposes, we could not find any published literature addressing the concurrent effect of FSP or OLP on the calf health and well-being.

Inconsistent with Wu et al. (2021), we observed no treatment effect on blood glucose and BUN concentrations which is in line with other studies (Kargar et al., 2018a,b; Selvi and Tapki, 2019). Supplementing FSP (10 g/d) to Mohabadi dairy goats from 30 d before to 30 d after kidding did not affect blood concentrations of glucose or BUN; however, there was a tendency for increased blood glucose and a significant reduction in blood BUN postkidding when FSP was supplemented (Yari et al., 2018). As anticipated, blood glucose concentration gradually tended to decrease with advancing calf age, showing a normal physiological fuel shift in the source of energy from glucose to short-chain fatty acids with the beginning of ruminal fermentation and development (Kargar et al., 2018a,b).

Cholesterol and triglycerides are 2 main indicators of lipid metabolism, and their concentrations are highly associated with health and diminished early mortality in calves (Renaud et al., 2018) as observed in the present study. Amad et al. (2013) reported a higher blood concentration of cholesterol but not triglycerides (no changes) when broiler chickens were supplemented with a phytogetic feed additive containing a mixture of essential oils with *trans*-anethole as one of the major active substances, and attributed this effect to improved lipid digestibility because of increased digestive and bile enzyme secretion. Wu et al. (2021) reported lower concentrations of blood cholesterol and triglycerides when a blend of oregano essential oils and cobalt-lactate (at 0, 4, and 7 g/d) was fed to fattening lambs; however, Selvi and Tapki (2019) reported no changes in blood concentrations of cholesterol and triglycerides when oregano essential oil was fed to milk-fed calves. Although the reason(s) for increased concentration of triglycerides, in the present study, is(are) unclear, our data do not support a triglyceride-lowering effect for FSP and OLP.

Concentrations of blood total proteins and albumin are longer-term indicators of AA metabolism and immune states, whereas BUN concentration may more accurately reflect the short-term dietary affects on rumen $\text{NH}_3\text{-N}$ formation and hepatic N turnover (NRC, 2001; Kargar et al., 2018b). As such, globulin and albumin:globulin ratio are used to assess the systemic immune function when metabolic demands are high (Kargar et al., 2018a,b). Consistent with others (Yari et al., 2018; Selvi and Tapki, 2019), the lack of FSP or OLP effects, in the present study, on blood concentrations of total

proteins, albumin, BUN, globulin, and albumin:globulin ratio may imply that no alterations in the demand and competition for AA occurred and, in turn, shows no challenge to hepatic synthesis of albumin or globulin, required to sustain nutrient transportation and normal immune function (Kargar et al., 2018a,b).

Alanine transferase (mainly in liver) and AST (in liver and other tissues such as skeletal muscles) are main indicators of tissue damage or dysfunction. In the present study, blood concentration of ALT was not affected by treatment; however, FSP versus OLP increased blood concentration of AST on d 30, indicating a transient tissue damage or dysfunction, which was subsequently ameliorated. Hajalizadeh et al. (2019) observed that FSP supplementation to fattening lambs had hepatoprotective effect and decreased blood concentrations of AST and ALT in a dose-responding manner. Selvi and Tapki (2019) also reported a reduction in concentration of AST when oregano essential oil was supplemented to milk-fed calves.

The phenolic compounds, such as flavonoids and phenolic acids, are mainly responsible for antioxidant activity of fennel seeds and oregano leaves (Badgujar et al., 2014; Alagawany et al., 2020). Inconsistent with others (Hashemipour et al., 2013; Ri et al., 2017) and at the dosages used in the present study, antioxidant activity of FSP or OLP was not translated into blood concentration of TAS or MDA. We propose that the potential antioxidant properties of FSP or OLP were spatially limited to affect the local gut health and feed efficiency or the function of the intestinal epithelium, not allowing detection of the likely effects in the blood; however, the positive long-lasting effects of this feeding strategy early in life was depicted on growth performance and animal health.

CONCLUSIONS

Supplementing FSP and OLP to the liquid feed of dairy calves during the first month of life improved their growth performance through increasing the feed intake or feed efficiency as well as mitigating the health-related abnormalities. These herbal constituents, especially FSP, were effective in reducing the rectal temperature as well as decreasing the chance of experiencing, and the duration of, diarrhea and BRD. As a new strategy, findings of the present study indicated the potential use of FSP and OLP as feed additives to improve the growth and health of calves.

ACKNOWLEDGMENTS

Authors thank Dam Gostaran-e-Kavir Animal Husbandry Farm (Yazd, Iran) for partial funding of this re-

search, providing suitable experimental conditions, and for diligent animal care. The authors have not stated any conflicts of interest.

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